GENOMICS LABORATORY REPORT

Next-Generation Sequencing & Clinical Bioinformatic Evaluation
Accredited ISO-15189 / CLIA High-Complexity Laboratory

ID: REP-2026-0003
Status: DELIVERED
Date: Oct 26, 2026

1. REQUISITION & INVESTIGATOR INFORMATION

Requester Name:	Dr. Priya Nair	**Requisition Title:**	Population Cohort Targeted Cancer Predisposition Panel
Organization / Dept:	Institute of Rare Genetic Diseases	**Sample Type:**	DNA
Request Code:	REQ-2026-0008	**Priority Level:**	ROUTINE

2. SPECIMEN IDENTIFICATION & QUALITY METRICS

Sample Barcode:	SMP-2026-0011
Collection Date:	1/8/2026
Received Date:	3/8/2026

QC PARAMETER	MEASURED VALUE	QC STATUS
Fluorometric Conc.:	88 ng/µL	**PASS**
Purity A260/280 & 230:	1.88 / 2.15	RIN/DIN: 9.2

3. SEQUENCING RUN SPECIFICATIONS

Run Code:	RUN-2026-0004	**Run Assay Type:**	WGS
Platform / Chemistry:	PacBio	**Read Configuration:**	HiFi 15-20kb
Target Coverage:	20x target depth	**Execution Status:**	COMPLETED

4. BIOINFORMATICS PIPELINE & INTERPRETATION SUMMARY

Pipeline: DeepVariant v1.6 + Clair3

Reference Assembly: GRCh38

Executive Findings & Summary:

Heterozygous pathogenic frameshift variant in BRCA2 c.5946delT (p.Ser1982ArgfsTer22) confirmed with 32x high-fidelity read support.

5. LABORATORY CERTIFICATION & SIGN-OFF

This test was developed and its performance characteristics determined by the Genomics Laboratory. It has been validated in accordance with laboratory accreditation standards for research and clinical oversight.

VERIFIED BY (LAB STAFF):
Marcus Brody
Date: 2026-10-26

APPROVED BY (LAB DIRECTOR):
Laboratory Medical Director, Ph.D., FACMG
Electronic Signature Verification: PASSED